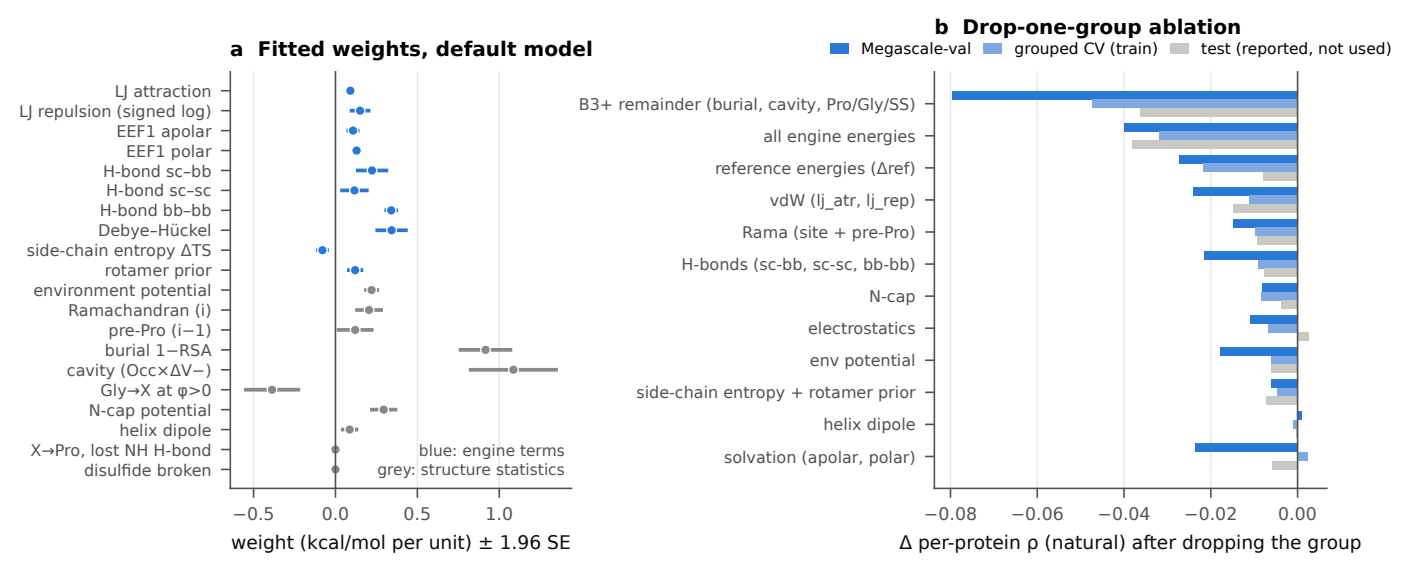
a Fitted weights, default model
LJ attraction
LJ repulsion (signed log)
EEF1 apolar
EEF1 polar
H-bond sc–bb
H-bond sc–sc
H-bond bb–bb
Debye–Hückel
side-chain entropy ΔTS
rotamer prior
environment potential
Ramachandran (i)
pre-Pro (i−1)
burial 1−RSA
cavity (Occ×ΔV−)
Gly→X at φ>0
N-cap potential
helix dipole
X→Pro, lost NH H-bond
disulfide broken
blue: engine terms
grey: structure statistics
−0.5
0.0
0.5
1.0
weight (kcal/mol per unit) ± 1.96 SE
b Drop-one-group ablation
Megascale-val
grouped CV (train)
test (reported, not used)
B3+ remainder (burial, cavity, Pro/Gly/SS)
all engine energies
reference energies (Δref)
vdW (lj_atr, lj_rep)
Rama (site + pre-Pro)
H-bonds (sc-bb, sc-sc, bb-bb)
N-cap
electrostatics
env potential
side-chain entropy + rotamer prior
helix dipole
solvation (apolar, polar)
−0.08
−0.06
−0.04
−0.02
0.00
Δ per-protein ρ (natural) after dropping the group